

Lukas Jander

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Education

Rochester Institute of Technology, GPA: 3.99/4.00, Dean's List, Honors Program Rochester, NY
BS/ME Mechanical Engineering Expected Graduation May, 2027

Experience

Laboratory for Laser Energetics

Engineering Intern

January, 2025 – July 2026

- Designed and analyzed a flexure-based 3 DOF mirror mount
 - Ansys structural and Sigfit simulations drove design choices based on optical requirements
 - Monte Carlo simulations aided in tolerancing requirements and design choices
- Simulated an argon gas cooling fixture utilizing conducting copper coils in liquid nitrogen
 - Ansys thermal simulations and hand calculations compared favorably with measured results
- Created a passive lift assist and positioning mechanism for a 35lb calorimeter
- Designed a rigging fixture for a 60lb X-Ray camera

Insulet Corporation

Quality Engineering Intern

January, 2026 – April 2026

- Wrote test procedures for cyclic load testing of a medical device
 - Programmed Instron tensile test methods to automate testing
- Created torsional test setups to evaluate repeatability of manufacturing procedures
- Characterized repeatability of surface roughness measurements to investigate ink adhesion issues

SEPAC Inc.

Engineering Intern

May, 2024 – August, 2024

- Designed a test setup to automate brake performance evaluation
- Wrote programmable logic controller (PLC) code to automate test procedures
- Performed hardware and software repairs on existing test setups to minimize downtime

Rochester Institute of Technology

Researcher in a Microelectronics Lab

May, 2023 – August, 2023

- Evaluated sensitivity of a MEMS viscosity sensor
- Designed a test setup to amplify and filter sensor output signals
- Used LabVIEW to create a user interface and data collection program
- Presented findings in *Waterproofing a Thermally-Actuated Vibrational MEMS Viscosity Sensor* and a research symposium

Projects

RIT Competitive Rocketry Team: Airbrakes Subteam Member

- Performed FEA to evaluate mechanical load cases experienced during flight
- Designed an active drag control mechanism to meet target apogee
 - Wrote MATLAB scripts to predict load on components and optimize design
- Performing CFD on a computing cluster to obtain realistic drag estimates

Combination Lock Solver

- Designed a device to open combination locks and display the solution
- Arduino controlled stepper motor, solenoid, and a beam break sensor drive the mechanism
- Testing of servo motor and design of PCB in progress

Skills

Software: ANSYS, MATLAB, NX, LabVIEW, SigFit, most CAD packages

Miscellaneous: GD&T proficient, Fluent in German and Czech

Leadership

Rochester Institute of Technology

Treasurer of Rock Climbing Club

May, 2023 – August, 2026

- Organized events, fundraised, and communicated with university staff about budget